Personalization in Public Transport Passenger Information Systems: A Systematic Review and Framework

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In recent years, efforts to promote public transport usage have emphasized the integration of personalization into passenger information systems. Whilst numerous studies have explored this topic, a comprehensive overview of the state-of-the-art remains absent. To address this gap, we systematically reviewed 91 research articles published between 2000 and October 2024. Based on our review, we introduce a comprehensive framework that organizes the field along three key dimensions: personalization objects, attributes, and evaluation. Furthermore, we provide valuable insights for researchers and practitioners engaged in the study, design, and development of personalized passenger information systems and identify research opportunities to guide future advancements in the field.

CCS Concepts: • **Human-centered computing → Interactive systems and tools**; • **General and reference → Surveys and overviews;**

Additional Key Words and Phrases: Passenger information systems, public transport, personalization, systematic literature review

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1 Introduction

Efforts to promote **public transport** (**PT**) are increasingly prioritized across regions to alleviate congestion and achieve sustainability goals. Notable European projects, such as MaaS4EU [45], MaaSiFiE [122], and iMove [16], have advanced **Mobility-as-a-Service** (**MaaS**) solutions aimed at enhancing travel convenience, reliability, and flexibility. These enhancements to the travel experience have proven effective in encouraging car owners to shift toward PT [134].

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Building on these advancements, numerous studies have explored approaches to improve the travel experience further. For example, expanding Copenhagen's metro network to improve availability resulted in a 3% reduction in car ownership [113]. Similarly, micro-navigation systems have been shown to enhance urban bus passengers' experiences [52]. Research has also highlighted the relationship between crowdedness, waiting-time reliability, and user satisfaction [151]. Furthermore, Drabicki et al. [42] modeled the effects of real-time crowding information, shedding light on the negative effects of overcrowding.

With the expansion of PT systems and the growing availability of advanced travel planners, **passenger information systems** (**PIS**) now provide passengers with abundant information. Unfortunately, a prevailing issue is cognitive overload [72], as travelers must process large volumes of data and integrate multiple sources for increasingly complex planning tasks [94]. One effective strategy to mitigate this challenge is personalization [100].

In recent years, personalization has gained considerable traction across diverse fields such as e-commerce (e.g., [168]), music streaming (e.g., [84, 129]), and gaming (e.g., [126, 159]). The reasons for its adoption are multifaceted [176]. From a human-centered perspective, personalization enhances user experience [86], tailors information and recommendations [147, 153], and boosts engagement [127, 154]. From an industry perspective, it increases conversion rates [160], influences user behaviour [40], and positively impacts sales [24]. In PT, personalization holds significant potential to improve the travel experience by streamlining journeys and making them more enjoyable. Several studies have explored personalization in PIS, resulting in many publications that could benefit the PT sector. However, two critical issues persist: (1) the existing body of literature is fragmented, lacking a cohesive, holistic overview, and (2) no clear indication of trends or research gaps to guide future work exists.

This article presents a comprehensive overview of research on personalization in **public transport passenger information systems** (**PT PIS**). It identifies, analyzes, and categorizes 91 articles, resulting in a framework highlighting key personalization elements and their applications. Additionally, it outlines six central research directions to guide future studies. These contributions offer a comprehensive overview of the various possibilities for personalization in PT PIS along with their added value. They aid researchers and practitioners in navigating the field while promoting a standardized approach to researching, designing, and developing personalized PT PIS.

The article is structured as follows: Section 2 covers the background, Section 3 details the sample characteristics, Section 4 presents the Personalization in PT PIS framework, Section 5 explains its practical use, and Section 6 highlights key insights and future research directions. The supplementary materials contain details on the research method.

2 Background

This section discusses reviews relevant to our topic and establishes a conceptual basis for personalization and PT PIS.

2.1 Related Reviews and Surveys

Several narrower-scope reviews have explored aspects of personalization in transport, mobility, and tourism. For instance, Tito et al. [156] reviewed the use of contextual information in bus PIS, but context is just one facet of personalization, and their review is restricted to bus-related applications. Similarly, Anagnostopoulou et al. [6] reviewed persuasive strategies to influence transportation behaviour, identifying personalization as one approach without broader exploration. A 2010 review [85] focused on methods for personalizing e-tourism recommendation systems. While they address personalization more comprehensively, it is outdated and limited to e-tourism and recommendations. Meanwhile, reviews by Nikitas et al. [117] and Jevinger et al.

[82] broadly examined the use of AI in transportation but lacked a focus on personalization. Additionally, Zhang et al. [179] conducted a focused review of route recommendation systems, providing limited insights into other personalization aspects. These prior reviews have not fully addressed the complex nature of personalization in PT PIS nor focused on the (entire) public transportation domain. To address these gaps, our review provides a comprehensive perspective on personalization in PT PIS, introduces a framework to guide research and practice, and identifies key directions for future advancements.

2.2 Definition of Personalization and Related Terminology

The definition of personalization varies across disciplines and research areas [49]. In information systems, several definitions have been proposed over time. For instance, early on, Hagen et al. [63] stated that *"personalization is the ability to provide content and services that are tailored to individuals based on knowledge about their preferences and behaviour"*. Blom [15] emphasized what changes are made and the rationale behind these changes, defining personalization as "*a process that changes the functionality, interface, information content, or distinctiveness of a system to increase its personal relevance to an individual"* (p. 313). Fan and Poole [49] adapted this to emphasize broader applicability: *"a process that changes the functionality, interface, information access and content, or distinctiveness of a system to increase its personal relevance to an individual or a category of individuals"* (p. 183). In the context of mobile devices, Asif and Krogstie [9] stressed user modeling and context-awareness: *"a controlled process of adaptation of a service to achieve a particular goal by utilizing the user model and the context of use"* (p. 345). Graus and Ferwerda [61] highlight the goals of personalization, defining it as *"personalization is performed by adapting aspects of systems to match individual users' needs in order to improve efficiency, effectiveness, and satisfaction"* (p. 3). These definitions emphasize key aspects of personalization, including the content being personalized (e.g., interface, functionality), the target audience (individual or group), the purpose (enhancing relevance, matching needs, increasing efficiency, improving satisfaction), and the data types used (knowledge-based, information-driven, context-specific). They highlight the importance of the adaptation process.

For this review, we define personalization in the context of PT PIS as *"The process of adapting a passenger information system to (a group of) travelers based on personal and contextual factors. This adaptation can be implicit and explicit and aims to achieve a particular goal"*. This definition is specifically tailored to the domain of PT and PIS.

Several terms are often used interchangeably with or alongside personalization. We will clarify these terms and their relationship to personalization to provide a clear conceptual foundation for this research.

Adaptation is frequently regarded as a mechanism to achieve personalization [49]. Adaptive systems dynamically adjust to the user and/or their context over time [125]. For example, a system might adapt its recommendations based on evolving user preferences, whether these changes are detected implicitly or specified explicitly.

Customization is another term often conflated with personalization, but it specifically involves allowing users to directly input their preferences. For instance, selecting preferred topics in a news application to tailor the content. In this sense, customization can be considered as a form of explicit or user-initiated personalization [15, 49].

Individualization is a technique that uses data about individuals to address their specific needs and preferences. To understand individualization, it is essential first to examine segmentation, which divides a target population into smaller groups based on shared characteristics like demographics, interests or preferences [12]. While segmentation supports tailoring experiences to groups, individualization focuses on adapting to the unique preferences of each individual.

Context-awareness, also referred to as situation-based adaptation, is frequently associated with personalization. According to Dey and Abowd [2] "*A system is context-aware if it uses context to provide relevant information and/or services to the user, where relevancy depends on the user's task.*". By incorporating real-time contextual information, such systems can deliver information, recommendations, or content that is better aligned with the user's current situation.

2.3 Definition of PIS for Public Transport

PT traditionally referred to *"buses, trains, and other forms of transport that are available to the public, charge set fares, and run on fixed routes"* [131]. However, the range of options has expanded to include on-demand services, such as shared cars and bikes. In this study, *PT* refers to all transportation modes available for public use, including buses, trains, metros, trams, taxis, ferries, airplanes, car-sharing, and bike-sharing.

PIS in PT, also referred to as Advanced Traveler Information Systems or PT Information Systems, integrates both static and dynamic information to help passengers effectively plan and use PT. These systems have evolved significantly, offering real-time information and supporting trip planning and decision-making during journeys. PT PIS initially focused on a single transportation mode (e.g., metro). However, there has been a shift towards multi-modal PIS, which provides a seamless door-to-door travel experience [37] and integrates bundles of mobility services [54]. In this context, MaaS was introduced in 2014 to offer a comprehensive range of mobility services to customers [69]. Hietanen [73] defined MaaS as a model that addresses users' transport needs via a single interface, combining various transport modes into a tailored mobility package. Since its inception, MaaS has gained global traction and become a focus of innovation agendas [16, 47, 122]. However, it continues to evolve, resulting in varying definitions. When combined with personalization, MaaS has the potential to greatly enhance travel convenience, reliability, and flexibility.

3 Dataset Characteristics and Technology Readiness Phase (TRP)

The final corpus analysis consisted of 91 articles, see Table 1. Of these, 45 were published in journals, 43 in conference proceedings, and 3 in book chapters. Publication venues varied widely, reflecting the interdisciplinary nature of the topic. The top three venues were (1) IEEE Transactions on Intelligent Transportation Systems (6 articles), (2) International Conference on Intelligent Transportation Systems (4 articles), and (3) Transportation Research Procedia (3 articles). Geographically, most articles were affiliated with European institutions (61), followed by Asia (22), North America (13), South America (4), Africa (4), and Oceania (3). While only a few were published before 2005, the volume has grown steadily, surging since 2015 (see Figure 1).

Table 3 shows the PT modes in our sample. We included a mode if it was explicitly mentioned or could be inferred from the content of tables, figures, datasets, or PT provider information. Most articles included buses, with about one-quarter focusing exclusively on buses without considering other modes. We observed similar patterns for other modes, with only 11 articles focusing solely on a single mode, such as trains [62, 80, 103], shared mobility [13, 142, 174], metros [11, 83, 98, 120], and airplanes [33]. Three articles discussed PT in general, not mentioning modes [19, 91, 173].

To assess the maturity of the technology in the reviewed articles, we classified them by their TRP (see Table 1). Initially developed by NASA and later standardized by the EU for broader application [102], the TRP differentiates research, development, and deployment phases (see Table 2), each divided into **technology readiness levels** (**TRLs**). Given the challenges in identifying exact TRLs, we classified them on TRP. Most articles were in the research phase, focusing on scenario formulation and requirements for PT PIS (e.g., [11, 89]). Articles exploring user preference acquisition from historical data for personalization were also categorized in this phase (e.g., [23, 96]). A

Table 1. An Overview of the 91 Articles Included in Our Sample, Including the Affiliation of the Authors in Terms of Country as well as the Identified TRP

Ref.	Lead author	Year	Cont.	Count.	TRP	Ref.	Lead author	Year	Cont.	Count.	TRP
[1]	Abdulrazak	2022	NA	CA	1	[91]	Kölker	2019	EU	DE	1
[4]	Al-Rahmaneh	2021	EU	ES	3	[92]	Komatani	2005	AS	JP	2
[5]	Alkhabass	2022	EU	SE/IT	2	[95]	Lathia	2011	EU	UK	1
[7]	Anagnostopoulou	2017	EU	GR	3	[96]	Lathia	2013	EU/NA	UK/US	1
[8]	Arentze	2013	EU	NL	1	[98]	Li	2023	AS	CH	2
[11]	Bajaj	2016	EU/AS	FR/IN	1	[101]	Liu	2022	AS	CN	3
[13]	Billhardt	2021	EU	ES	1	[103]	Lu	2017	AS	CN	1
[17]	Boero	2016	EU	ES/IT/CZ	3	[104]	Marcussen	2012	EU	NO	3
[18]	Borodinov	2019	EU	RU	1	[107]	Mnasser	2013	AF/EU	TN/FR	1
[19]	Bravo	2018	SA/EU	EC/ES	2	[108]	Mo	2021	NA/AS	US/CH	1
[20]	Brossard	2011	EU	FR	3	[110]	Mohan	2019	NA	US	1
[22]	Carmien	2005	NA	US	2	[111]	Mohan	2019	NA	US	2
[23]	Ceder	2020	AS/OC/EU	IL/NZ/DK	1	[112]	Moussa	2013	AF/EU	TU/FR	2
[25]	Chavhan	2020	SA/AS/EU	BR/IN/PT	2	[114]	Nakamura	2015	AS	JP	2
[26]	Cheikh	2010	EU	FR	2	[115]	Nakamura	2016	AS	JP	3
[27]	Chen	2023	AS/OC	CH/AU	2	[116]	Namoun	2021	AS/EU	SA/UK	3
[29]	Chow	2016	AS	CH	2	[120]	Noursalehi	2021	NA	US	1
[30]	Christmann	2008	EU	DE	1	[121]	Nuzzolo	2016	EU	IT	1
[33]	Darvishy	2008	EU	CH	1	[123]	Panou	2012	EU	GR	2
[34]	de Oliveira Tito	2014	SA	BR	2	[124]	Panou	2010	EU	GR	1
[35]	Dibbelt	2015	EU	DE	1	[128]	Park	2023	NA/AS	US/KR	2
[38]	Dinko	2023	EU	LV	1	[132]	Pressl	2010	EU	AT/DE	1
[43]	Eryilmaz	2014	EU	DE	1	[133]	Qoradi	2021	AS	SA	3
[44]	Essayeh	2017	EU	FR	2	[135]	Rehrl	2007	EU	AT/DE	3
[46]	Esztergar-Kiss	2021	EU	HU	3	[137]	Roor	2015	EU	DE	1
[51]	Ferreira	2018	EU	PT	2	[140]	Said	2018	AF/EU	EG/FR	2
[53]	Földes	2015	EU	HU	1	[142]	Samsel	2016	EU	DE	2
[56]	Garcia	2012	EU	ES	3	[143]	Samsel	2018	EU	DE	2
[57]	Gault	2019	EU	UK	3	[144]	Sanjust	2015	EU	IT	3
[58]	Gavalas	2020	EU	GR	3	[145]	Schlingensiepen	2018	EU	DE/RO	1
[60]	Goto	2002	AS	JP	3	[149]	Solar	2012	EU	ES	1
[62]	Griol	2013	EU	ES	2	[150]	Song	2018	NA	US	2
[64]	Handte	2016	EU	DE/UK/NL	3	[152]	Stanciu	2015	EU	RO	3
[65]	Hao	2024	AS	CH	2	[157]	Titov	2023	EU	DE	1
[66]	Haqqani	2017	OC	AU	1	[158]	Titov	2020	EU	DE	2
[67]	He	2022	AS	SG/CH	2	[162]	Turno	2023	EU	LV	2
[68]	Heckmann	2022	EU	DE	2	[163]	Twahirwa	2022	AF/AS	RW/IN	3
[71]	Herzog	2017	EU	DE	2	[169]	Vieira	2011	SA	BR	2
[75]	Hoar	2010	NA	CA	3	[170]	Vo	2015	AS	SG	2
[78]	Irani	2006	NA	CA	2	[173]	Wu	2022	NA/EU	US/DE/SE	1
[80]	Jevinger	2023	EU	SE	2	[174]	Yan	2024	EU	IR	1
[81]	Jevinger	2019	EU	SE	1	[177]	Zenker	2010	EU	DE	1
[83]	Jin	2017	AS	CH	1	[178]	Zhang	2011	NA	US	3
[87]	Katona	2017	EU	HU	1	[180]	Zhao	2021	AS	CH	2
[88]	Keller	2019	EU	DE	1	[181]	Zografos	2009	EU	GR	3
[89]	Keller	2020	EU	DE	1						

third of the articles were in the development phase, testing prototypes in experimental or simulated environments (e.g., [143, 158]). Finally, articles in the deployment phase focussed on deployed systems and, mostly, evaluated in operational environments (e.g., [133, 152]). TRP classifications showed no notable variation across years, continents, or countries.

4 Framework of Personalization in PT PIS

The reviewed articles present diverse ideas and applications for personalized PT PIS. To clarify their interconnections, we developed a framework based on the sample, providing an overview to guide future personalized PT PIS research and development. We structure the framework around three dimensions: *personalization objects*, *attributes*, and *evaluation* (see Figure 2).

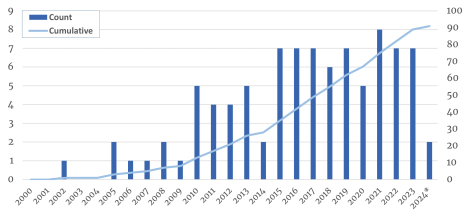

Fig. 1. Publication date of articles in the sample. The bars present the number of articles for that specific year. The line chart presents the cumulative growth of articles in the period 2000-October 2024.

Table 2. Description of the TRPs and Their TRLs

TRP	TRL	
TRP 1	TRL 1	Basic principles observed
Research	TRL 2	Technology concept formulated
	TRL 3	Experimental Proof of Concept
TRP 2	TRL 4	Technology validated in the lab
Development	TRL 5	Technology validated in relevant environment
	TRL 6	Technology demonstrated in relevant environment
TRP 3	TRL 7	System model/prototype demonstration in operational environment
Deployment	TRL 8	System complete and qualified
	TRL 9	Actual system proven in operational environment

Table 3. Transportation Modes and Number (#) of Related Articles in the Sample

Mode	#	References
Airplane	6	[33, 35, 56, 137, 149, 181]
Bus	72	[1, 4, 5, 7, 8, 17, 18, 20, 22, 23, 25–27, 29, 30, 34, 35, 38, 43, 44, 46, 51, 53, 56–58, 60, 64–68, 71, 75, 78, 81, 89, 92, 95, 101, 104, 107, 108, 110–112, 114–116, 121, 123, 124, 128, 132, 133, 135, 137, 140, 142, 144, 145, 149, 152, 158, 162, 163, 169, 170, 177, 178, 180, 181]
Ferry	3	[23, 56, 181]
Metro	38	[4, 11, 17, 20, 23, 25, 35, 44, 46, 53, 65, 67, 68, 71, 83, 87, 95, 96, 98, 103, 107, 108, 110–112, 116, 120, 121, 135, 137, 140, 142, 144, 145, 150, 152, 180, 181]
Shared mobility	16	[7, 13, 17, 43, 46, 58, 66, 68, 87, 89, 108, 110, 142, 143, 158, 174]
Taxi	6	[4, 17, 30, 101, 108, 110]
Train	39	[4, 5, 8, 17, 20, 23, 26, 35, 44, 46, 56, 60, 62, 66, 68, 71, 80, 81, 88, 89, 95, 96, 107, 108, 110–112, 114, 121, 135, 137, 140, 142, 144, 149, 150, 157, 158, 178]
Tram	31	[4, 8, 17, 20, 25, 26, 38, 43, 44, 46, 53, 58, 68, 71, 87, 88, 107, 108, 110, 112, 116, 123, 124, 132, 135, 137, 144, 145, 152, 158, 181]

4.1 Dimension I: Object of Personalization in Personalized PT PIS

The personalization object—the "what" that researchers aim to personalize—includes various elements (see Figure 2). Around half the studies explored multiple aspects, leading to their classification across different categories (see Table 4).

4.1.1 Recommendations. The most common personalization object is recommendations, which emerged as an increasingly popular research area and also gained popularity in PT PIS. Recommendations aim at assisting users in their search and decision-making processes, enhancing the overall user experience [3, 21]. As shown in Table 4, the literature explores various types, including route, destination, and people recommendations.

Route recommendation provides travelers with PT routes tailored to personal and contextual factors. Personalized route recommendations enhance traveler satisfaction [26, 64, 107, 173], especially for those with special needs [19, 60], and promote sustainable travel by emphasizing eco-friendly options or simplifying the planning of such routes [46, 68, 111, 140]. Moreover, they help manage routing requests efficiently while optimizing traffic flow [8, 116, 173]. Zhang et al. [179] present a review focussing solely on route recommendation in PT. Within our sample, the focus varied depending on the TRP. Some articles analyze relationships between attributes and route choices (e.g., [88, 110]), while others focus on learning user preferences (e.g., [8, 123]). Several studies developed route recommendation models, broadly divided into route *optimization* and *classification* models.

Route *optimization* methods identify the best route within PT networks based on user queries, typically an **origin-destination** (**OD**) pair. These methods often evaluate criteria such as distance, time, cost, and preferences. Search-based algorithms, such as Dijkstra's algorithm [36], systematically explore all possible paths to identify the optimal solution. These methods can be enhanced by incorporating personalized optimization constraints [177], such as weighted costs based on user preferences (e.g., [23, 35]), or special needs like disabilities (e.g., [19]). Although reliable, these

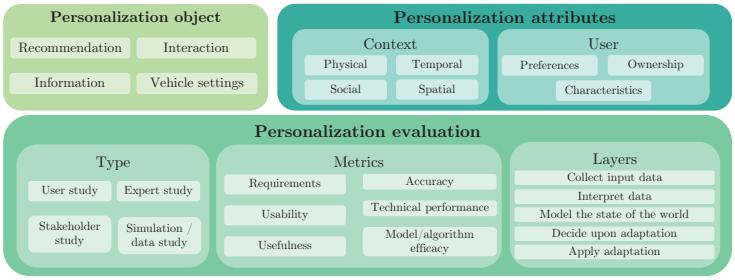

Fig. 2. Framework for personalization in PT PIS. The visualization illustrates the three dimensions of the framework: *personalization object*, *attributes,* and *evaluation*, and their identified categories and subcategories.

Table 4. Objects of Personalization in the Articles in the Sample, Per (sub) Category

Category	#	Subcategory	#	References
Recommendation	66	Route	63	[1, 4, 5, 7, 8, 11, 17–19, 23, 25–27, 29, 33–35, 38, 43, 44, 46, 53, 58, 60, 64–68, 71, 80, 81, 83, 87, 88, 95, 98, 101, 103, 110–112, 114–116, 121, 124, 128, 132, 135, 140, 142–145, 149, 150, 157, 162, 170, 173, 174, 177, 180]
		Destination	4	[13, 58, 174, 177]
		People	2	[27, 137]
Information	41	Travel time	26	[1, 7, 26, 29, 30, 33, 38, 51, 56, 57, 62, 75, 88, 89, 92, 96, 104, 108, 124, 144, 145, 152, 163, 169, 178, 181]
		Vehicle	23	[7, 22, 29, 38, 51, 56, 58, 62, 75, 81, 88, 89, 92, 104, 120, 124, 133, 144, 152, 163, 169, 178, 181]
		Navigation	18	[1, 22, 29, 30, 33, 51, 60, 64, 89, 92, 135, 143–145, 169, 177, 178, 181]
		Disturbances	17	[1, 22, 26, 27, 33, 38, 43, 51, 56, 57, 60, 81, 88, 89, 104, 135, 181]
		Fares	7	[27, 30, 38, 43, 62, 143, 144]
		Facilities	4	[20, 30, 60, 144]
		Unspecified	2	[132, 158]
Interaction	31	Communication modality	17	[19, 22, 26, 29, 43, 51, 57, 60, 64, 81, 88, 133, 143, 145, 149, 152, 181]
		User interface	8	[20, 22, 75, 78, 88, 91, 107, 145]
		Content presentation	5	[7, 29, 92, 107, 140]
Vehicle settings	1	Volume infotainment	1	[143]
		Climate control	1	[143]
		Audio Source	1	[143]

models require pruning for computational efficiency. Probabilistic methods use heuristics and approximations to find satisfactory solutions. For example, stochastic networks can manage uncertainty [55, 118], such as delays or disruptions [18, 170], enabling more realistic and reliable recommendations. Inspired by social behaviours (e.g., ant colonies, bee hives [41]), swarm intelligence addresses complex optimization problems. These approaches can incorporate travelers' historical behaviour data, adapting based on past experiences or knowledge [87]. **Machine learning** (**ML**), particularly reinforcement learning, enhances dynamic adjustment and reduces computation time [67].

These models are robust and scalable, yet uncertainty and computational demands present challenges. Personalization occurs during route generation. While objective metrics such as cost, CO2 or accessibility metrics [128] can be implemented as optimization parameters, subjective personalization factors should be added as weighted constraints. Moreover, they require ML augmentations to learn and update user preferences.

The second personalization approach is *classification*, where models select the best route from pre-generated options. These options are created in a pre-processing phase via conventional optimization techniques [101] or external services such as OpenStreetPlanner or Google Maps (available in Table 2 in the supplementary material[1]). Personalization occurs during the route selection phase. Methods used can be categorised into collaborative filtering, content-based filtering, and knowledge-based filtering.

[1]Tabel 2 in the supplementary material details identified open datasets relevant to personalized PT PIS.

In **collaborative filtering** (**CF**), models recommend routes based on the travel history or ratings of similar users [11, 98]. This approach proves effective in providing recommendations during peak hours [98]. **Content-based** (**CB**) recommendations use route attributes and user profiles, learning preferences from historical interactions and route choice data (e.g., [8, 112, 114]). For example, utility-based models select the best route based on factors such as travel time, cost, and transport modes [111]. Both CF and CB methods face challenges such as the cold start problem—lacking data for new users or routes—and data sparsity, where limited user interactions hinder recommendations. Knowledge-based models, which rely on explicitly defined criteria like user preferences or contextual information, are less affected by these issues (e.g., [29, 34, 112, 140]). To improve recommendation quality and address these limitations, hybrid models combine approaches [11, 71], integrating user history, explicit preferences, and crowd-sourced data [101].

The choice of approach depends on data availability and system goals. Optimization algorithms are well suited for straightforward scenarios focusing on objective attributes such as travel time, cost, or distance, and when data on personalization attributes is limited. In contrast, classification models may be more suitable for more intricate, subjective personalization that emphasises user experience. In both cases, careful system design is essential to balance scalability, support growing user bases or datasets, ensure effectiveness and efficiency, and maintain real-time performance.

Additionally, some research explores **destination recommendations**, such as tailoring suggestions for pickup/drop-off charging stations [13, 58] or **points of interest** (**PoIs**) like music events [177]. Destination recommendations are more common in tourism systems [136], suggesting leisure activities, dining options, or recreational spots. However, they are less prevalent in PT PIS, where personalization tends to focus on practical, route-based guidance. In PT PIS, destination recommendations typically rely on content-based approaches, utilizing historical data on frequently visited locations.

Little research explores **people recommendations** [27, 137] within the PT PIS context, aiming to connect passengers who are either travelling on or waiting for the same transport service. This approach fosters social connections by using a knowledge-based method that combines geolocation, shared routes, and user preferences. However, it also raises significant privacy concerns, as the system requires detailed tracking of social relationships [137].

An interesting observation is the limited use of **explanations** to clarify *why* a specific recommendation was made [71, 95, 123]. While various explanation methods have been widely studied in general recommender system research [155], their integration into PT PIS remains sparse. Only three articles incorporated explanations [71, 95, 123], and none evaluated whether these explanations were effective or aligned with the recommendations provided. This gap highlights a potential area for improvement in enhancing user trust and transparency within personalized PT PIS.

4.1.2 Information. Personalized information is the second most prominent object of personalization in PT PIS. Unlike recommendations, which guide users in decision-making, personalized information enhances the user's understanding by adapting content that is not necessarily prescriptive. Table 4 shows the types of adapted information: information about **travel time**, **vehicles**, **navigation**, **disturbances**, **facilities**, and **fares**. While some information types are closely related, articles do not always include personalized information on related types; thus, we treat them separately.

Travel time is highly valued in PT [171] and has been associated with traveler stress [48]. Personalized travel time information encompasses different aspects but often relies on real-time data. Studies have explored personalizing information about the schedule (e.g., [51, 56, 96, 169]), journey durations (e.g., [7, 30, 124]) and the timing of upcoming vehicles relevant to the traveler's journey. For example, Hoar [75] developed a system providing real-time arrival estimates to inform

passengers whether they need to walk, run, or delay their departure to catch their bus. Overall, studies show promising results, with a strong emphasis on the accuracy of real-time predictions (e.g., [51, 169]).

Personalized **vehicle** information often complements real-time schedule updates by providing real-time location updates for vehicles relevant to the traveler's journey (e.g., [104, 132, 133]). This empowers travelers to make better-informed decisions, such as when to depart or whether to adjust their route. In addition to location updates, personalization in vehicle-related information includes occupancy or seat availability [120, 163, 181]), vehicle type (e.g., the train-type as presented by [62]), and CO2 emissions (e.g., [7, 144]). Furthermore, personalized vehicle accessibility information is provided for individuals with specific accessibility needs (e.g., [124]), promoting inclusivity in PT PIS.

Personalizing **navigation** information primarily addresses general navigation, such as helping travellers identify routes and directions. However, several studies explore more fine-grained micro-navigation, addressing specific tasks, such as identifying where to go inside a vehicle [89] or inside stations and other buildings [33, 60, 89, 135]. These detailed navigation aids enhance the traveler experience [64], especially for those with additional needs. However, implementing such solutions often relies on advanced technologies like beacons, augmented reality, or physical prototypes. As a result, we observe little implementations of these solutions in practice. Additionally, travelers receive support through alerts or notifications, such as warnings when they head in the wrong direction [60, 64] or reminders about when and where to board or alight a vehicle [22, 51, 64, 145]. These features are particularly beneficial for infrequent or unfamiliar travellers, individuals with special needs, or those in unexpected situations. However, notifications can disrupt memory and attention [77], particularly for experienced travellers. This highlights the importance of careful design and evaluation to ensure notifications' effectiveness while minimizing interruptions—an aspect often overlooked in existing studies. A noteworthy example of accessibility-focused personalization is Abdulrazak's work, tailoring route accessibility information to older adults' needs, considering factors like slopes and resting areas [1].

During PT **disruptions**, travellers often express an increased need for information, both in advance to prepare for potential issues and during the event to adapt to evolving circumstances [31, 167], particularly at an individual level [76]. Many studies focus on personalized real-time updates on delays, cancellations, and other changes affecting specific journeys. Researchers commonly use alerts and notifications to proactively inform passengers of relevant disruptions [51, 57, 181]. These personalized updates can foster a sense of control during disruptions, an aspect often lacking in such situations [57]. However, the accuracy and reliability of these updates are critical. Incomplete or incorrect information may undermine trust in PT PIS and negatively affect perceptions [130]. Ferreira et al. [51] emphasises the role of social accountability by incorporating user feedback and ratings into their system. This approach holds operators accountable for delays and deviations from timetables, creating social pressure to improve service reliability.

Personalized **fare and ticket** information offers travelers tailored insights into the cost of their journey, dynamically adapting to any changes. Personalized information about **facilities** enhances the overall travel experience by addressing higher-level traveler needs and offering tailored support [165]. This type of information spans a variety of services, including shops, ticket counters, restaurants, cafes, and cultural institutions [20, 30, 60, 144].

4.1.3 Interaction. Personalized interaction can enhance the user experience in PT PIS, particularly for travelers with diverse preferences or impairments that may affect their ability to interact with the system [10]. Research in this area primarily focuses on personalizing communication modalities, user interfaces, and content presentation.

Personalized **communication modalities** adapt how systems deliver information to travelers. A common approach is adapting the timing of push notifications (e.g., [43, 51, 133, 181]). Traditional PT PIS typically provide information reactively when users seek it. In contrast, personalized notifications can proactively assist users during planning and journeys as long as they are timely, relevant, and minimally disruptive [181]. Systems can also adapt communication dynamically based on travelers' situational needs and abilities [60]. For example, hands-free spoken instructions and haptic navigational prompts are beneficial during journeys (e.g., [152, 181]), and wearable devices that deliver alerts, such as notifications to accelerate walking-pace to make a connection, further illustrate the utility of personalized modalities [29]. However, determining the most suitable communication format for each situation remains challenging, as different modalities may be effective in different situations [91]. Personalized communication modalities are also valuable in supporting travelers with impairments and promoting information equity. For example, text-to-speech engines and multi-modal outputs combining spoken, visual, or textual information assist those with visual or hearing impairments, enhancing accessibility and reducing uncertainty [19, 64]. One study explored innovative solutions such as RFID tags, scannable markers, wearable gadgets, and augmented reality to aid travelers [145]. However, accessibility and feasibility limitations and the immature state of the technology (TRP 1) may hinder widespread implementation and adoption.

When personalizing **user interfaces**, the primary focus is on adapting graphical elements such as shape, location, or visibility. Some methods involve generating entire user interfaces tailored to user needs (e.g., [20, 91]), while others personalize specific user interface elements (e.g., [75, 107]). These adaptations improve usability in various contexts, ensuring that travelers have better access to relevant content. Given the sheer volume of information offered by current PT PIS, users may experience information overload. Tailoring the information format helps mitigate this issue, allowing travelers to quickly identify and engage with the content most relevant to their needs [72, 97].

Personalizing **content presentation** tailors how systems deliver information to the user. For instance, some studies focus on the wording and style of messages [7] or adapting information presentation to align with the user's ability [107]. Personalized spoken interactions are also explored, such as adjusting dialogue styles by modifying turn lengths or including follow-up questions to create more natural, user-friendly exchanges [62, 92]. Recent advancements in **large language models** (**LLMs**) and generative AI provide new opportunities for personalized content presentation. These technologies enable PT PIS to offer conversational assistance in natural language, either through text or spoken dialogue (e.g., [109]). However, their implementation faces challenges: the computational demands of LLMs may restrict their scalability, and issues like "hallucinations"—erroneous outputs generated by AI—could reduce user trust and hinder widespread adoption.

4.1.4 Vehicle Settings. Personalizing **vehicle settings** is an untapped opportunity within PT PIS. Whilst researchers have extensively explored this area for private vehicles (e.g., [28, 138]), its potential for shared vehicles remains largely unexplored. Only one study discusses the personalization of preferred vehicle settings [143]—including infotainment, climate control, and seat or mirror positions—with a mobile app profile to improve convenience and comfort for travelers. However, the practical implementation of such systems remains unrealized. Further research is needed to evaluate the feasibility, reliability, and user acceptance of such systems in shared mobility contexts [143].

4.2 Dimension II: Personalization Attributes in Personalized PT PIS

In the context of PT PIS, research uses various attributes to personalize recommendations, information, interactions, or vehicle settings. We categorize these attributes into two main types: context-related and user-related attributes. Each of these categories encompasses distinct

Table 5. Count of Identified Attributes Considered for Personalization Per TRP and Total (#)

Context		TRP 1	2	3	#	References
Physical	Infrastructure	7	10	5	22	[1, 4, 19, 22, 34, 38, 53, 65, 68, 75, 89, 116, 120, 124, 128, 135, 142, 145, 162, 163, 169, 180]
	Indoor/outdoor climate	12	12	6	30	[1, 4, 5, 7, 11, 17, 25, 26, 30, 34, 38, 58, 62, 68, 81, 89, 91, 101, 108, 110, 124, 128, 142, 143, 145, 149, 158, 163, 169, 174]
	Facilities	13	7	6	26	[4, 11, 13, 17, 19, 20, 27, 30, 38, 53, 56, 60, 68, 81, 88, 89, 98, 110, 114, 124, 132, 135, 137, 149, 162, 174]
Temporal	Current time and date	12	16	15	43	[1, 5, 11, 17, 20, 22, 29, 30, 34, 43, 44, 51, 53, 56–58, 60, 62, 64, 78, 80, 88, 89, 101, 104, 107, 114–116, 121, 143, 145, 149, 152, 158, 162, 163, 169, 170, 174, 177, 178, 181]
	Journey timing	27	27	19	73	[1, 4, 5, 7, 8, 11, 13, 17–20, 22, 23, 25–27, 29, 30, 34, 35, 43, 46, 51, 53, 56–58, 60, 62, 64–68, 78, 80, 81, 87–89, 96, 98, 103, 104, 108, 110, 112, 114–116, 120, 128, 132, 133, 135, 137, 140, 142–145, 149, 152, 158, 162, 163, 169, 170, 173, 174, 177, 178, 180, 181]
	Day of week	4	8	4	16	[20, 23, 27, 51, 78, 98, 101, 104, 108, 120, 140, 145, 152, 170, 174, 180]
	Time of day	10	9	3	22	[20, 23, 25, 30, 34, 35, 51, 78, 89, 96, 98, 101, 104, 107, 108, 110, 120, 140, 142, 145, 170, 180]
	Event, holiday, season	5	5	3	13	[4, 5, 20, 27, 51, 81, 98, 101, 107, 108, 120, 145, 158]
Spatial	Current location	14	13	11	38	[7, 13, 20, 22, 25, 29, 30, 33, 34, 44, 51, 53, 58, 60, 62, 64, 68, 80, 81, 88, 89, 104, 107, 108, 116, 133, 135, 137, 145, 149, 152, 158, 162, 169, 173, 174, 177, 178]
	Other locations	20	23	18	61	[1, 4, 5, 7, 13, 19, 20, 22, 25, 29, 30, 33, 34, 43, 44, 46, 51, 53, 56, 57, 60, 62, 64–68, 75, 80, 81, 87–89, 98, 101, 104, 107, 108, 111, 112, 116, 120, 123, 124, 128, 132, 133, 135, 137, 142, 145, 152, 162, 163, 169, 173, 174, 177, 178, 180, 181]
Social	Travel company	1	0	0	1	[110]
	Staff	2	1	1	4	[25, 53, 56, 145]
	Passer-by	8	5	2	15	[1, 11, 64, 65, 68, 81, 89, 91, 98, 103, 120, 137, 158, 163, 180]
User		**TRP 1**	**2**	**3**	**#**	**References**
Characteristics	Demographics	8	10	8	26	[1, 4, 17, 19, 20, 25, 26, 34, 38, 44, 46, 56, 60, 62, 67, 83, 101, 107, 108, 111, 124, 133, 142, 149, 158, 173]
	Special needs	12	10	5	27	[1, 4, 17–19, 22, 26, 29, 30, 34, 44, 53, 60, 62, 64, 81, 87, 89, 107, 114, 124, 132, 135, 145, 149, 158, 169]
	Knowledge level	4	4	0	8	[27, 30, 62, 81, 83, 92, 110, 111]
Preferences	Communication	3	5	4	12	[44, 57, 62, 64, 78, 89, 91, 92, 104, 149, 158, 181]
	Mode	14	11	8	32	[1, 5, 7, 8, 20, 26, 30, 35, 43, 44, 46, 53, 62, 64, 68, 71, 81, 89, 95, 101, 107, 110, 111, 116, 123, 135, 142–144, 150, 157, 173]
	Interchanges	13	9	2	21	[8, 11, 18, 23, 29, 34, 53, 65–68, 71, 81, 83, 87, 98, 101, 114, 123, 132, 135, 149, 157, 177]
	POIs and routes	18	19	8	41	[4, 5, 8, 18, 19, 26, 30, 34, 43, 44, 51, 60, 62, 67, 71, 75, 83, 87–89, 95, 96, 98, 101, 107, 108, 110–112, 114–116, 120, 121, 123, 128, 135, 149, 150, 162, 174, 177, 180]
	Time and day	20	15	7	39	[8, 11, 23, 29, 30, 34, 44, 46, 53, 57, 62, 64, 66, 67, 71, 81, 83, 88, 92, 95, 96, 98, 101, 107, 108, 110, 111, 114, 116, 121, 123, 128, 132, 135, 144, 145, 149, 150, 157, 158, 177, 180]
	Walking/cycling distance	11	6	7	19	[1, 7, 8, 11, 18, 29, 34, 44, 53, 64, 66, 67, 81, 107, 114, 123, 132, 135, 157, 177]
	Other people	7	1	0	8	[1, 5, 11, 81, 83, 137, 145, 177]
Ownership	Tickets/subscription	4	2	1	7	[30, 46, 89, 95, 111, 149, 158]
	Private vehicle	1	2	3	6	[4, 46, 68, 81, 111, 144]
	Personal items	3	3	0	6	[19, 26, 62, 89, 149, 177]

subcategories. Context-related attributes refer to factors that influence personalization based on the environment or situation. These are further divided into: *physical*, *social*, *temporal*, and *spatial context*. User-related attributes are centred around individual characteristics and preferences. These can be grouped into: *preferences*, *characteristics*, and *ownership*. Due to the variations in TRPs, attributes may be described in terms of plans, requirements, or prototypes rather than as actively implemented features. Despite this, we analyzed attributes across all TRPs to distinguish between intentions expressed in the early phases (TRP 1 and TRP 2) and those realized during the deployment phase (TRP 3).[2] Table 5 provides a detailed breakdown of each main and subcategory, including their total counts and distributions across the TRP.

4.2.1 Physical Context. Attributes related to the physical context are associated with the tangible environment of the PT system, which we divide into three categories. **Infrastructure** involves attributes tied to the physical structure of the PT system, often used to address travelers' accessibility needs. Improving accessibility facilitates individual mobility and contributes to social

[2]In some cases, assessing whether attributes were actually used for personalization was challenging. To address this, we included all attributes that could potentially contribute to personalization.

and societal inclusion [79, 141]. Key factors include vehicle accessibility, such as ease of entry and wheelchair-friendliness (e.g., [75, 124, 145]), and walkway accessibility such as slopes, steps, texture and safety of sidewalks or platforms (e.g., [19, 124]). Vehicle capacity helps to predict seat availability or vehicle crowdedness, aiding travelers' decisions (e.g., [34, 120]). Despite their importance, accessing infrastructure-related data remains challenging. Access to this data often relies on ontologies or APIs [19, 89], and limitations in data availability and standardization highlight the need for systematic efforts to document, quantify, and standardize such data [68, 145].

Some research uses **indoor and outdoor climate** to personalize PT PIS, with weather being a key attribute. Research typically represents it in two ways: (1) generalized categories like "sunny" or "rainy" [25], or (2) specific metrics such as temperature, precipitation, and air quality (e.g., [5, 142]). Balancing the granularity of weather APIs with practical relevance is key, as simpler classifications like "windy" are often more helpful than precise measurements such as exact wind strength [101, 163]. Other climate-related attributes, such as light levels and soundscapes, require sensor-based data collection, presenting logistical challenges. Consequently, studies often remain at lower TRP levels (e.g., [11, 30, 89]).

Attributes related to **facilities** often include hospitality-related factors, such as the availability of shops, cafes, and restaurants, which aid micro-navigation [60] or serve as ways to pass time [20, 30]. Additionally, accessibility features like seat availability, elevators, or access ramps support personalized suggestions (e.g., [17, 53, 149]). Most articles remain conceptual (TRP 1) with data availability and implementation being challenging. However, open-source platforms may have improved the availability of such data in recent years (see Table 2 in supplementary material).

4.2.2 Social Context. Attributes related to the social context primarily concern the presence and behaviour of others in the public transportation environment, which can be divided into three categories (see Table 5). Among these, **passer-by** was the most frequently addressed, focusing on the presence of others, crowdedness, and queues (e.g., [64, 120, 163, 180]). Studies in TRP 1 mainly consider crowdedness in terms of algorithm design or system requirements, with limited attention to real-time implications. At higher TRP levels, research uses historical data to predict crowdedness [180], while real-time monitoring methods using Wi-Fi signals [64] or cameras [163] allow dynamic adjustments. Historical data enables long-term trend analysis, such as forecasting crowdedness during peak hours. However, irregular events pose challenges, risking passenger trust if predictions are incorrect [120]. In contrast, real-time methods provide more accurate information but raise privacy concerns regarding surveillance, which were notably unaddressed in the reviewed studies. This gap highlights the need to explore the tradeoff between accuracy, trust, and privacy, as debated in broader IoT and video surveillance literature [119, 139].

Staff—including support personnel, drivers, and security—emerged in the reviewed studies (e.g., [25, 53, 145]). Staff members are important for assisting individuals with special needs or supporting passengers during disruptions. A notable implementation considers the dynamic state and behaviour of staff [25], integrating both current and historical data such as health condition, skills, fatigue, punctuality, and experience. However, they manually input these attributes, raising concerns about scalability and feasibility in practice.

Travel company emphasizes the impact of accompanying individuals, for example, on mode choice [110]. Although this remains underexplored in the PT domain, research can use insights from the tourism domain, where group recommender systems commonly account for travel companions [32]. Integrating such systems could enhance the travel experience, particularly for groups with diverse needs, such as those travelling with children or elderly.

4.2.3 Temporal Context. Research widely addresses attributes related to the temporal context, falling into five subcategories: current time and date, journey timing, day of the week, time of day,

and events, holidays, and seasons. Unsurprisingly, **journey timing** emerged as a key aspect of PT PIS, as it directly relates to network operations. This attribute concerns scheduled departure and arrival times, as well as actual travel times, which can be affected by delays, disruptions, and interchanges. As such, journey timing combines both static (e.g., timetables) and dynamic elements (e.g., real-time delays). Various open-source datasets (Table 2 in supplementary material) provide static timetable information, which is often combined with **current time and date** to enable real-time system adaptations. However, research does not always clearly articulate the effective use of these attributes in models. They often combine them with schedules to provide users real-time updates or comparisons of their remaining time before departure or arrival (e.g., [29, 64]).

Despite their importance, researchers underexplored the connections between journey time and other time-related attributes, especially during the development phase. Attributes like **time of day** and **day of week** are often readily available and play a critical role in enhancing travel time estimates [30, 35]. However, personalizing trip estimation algorithms using temporal data presents challenges. Temporal variations, like delays or disruptions, introduce complexities into the personalization process [25, 104]. Stochastic modeling offers one solution, using historical data to account for variability [170]. Similarly, Lathia et al. [96] propose using a geometric mean to create more robust estimates.

The influence of **events, holidays, and seasons** should be considered for personalized PT PIS [81], as these factors significantly affect travel patterns [5, 101, 108]. However, current research insufficiently addresses the methods for acquiring such data, and they do not use publicly available datasets. A notable exception is the approach described in [4], where bots automatically collect and upload event data from municipal services, ensuring consistent data availability without requiring manual intervention. In modeling, events are sometimes represented in a binary format [120], but this simplified approach may fail to capture the nuanced ways events influence travel behaviour.

4.2.4 Spatial Context. The spatial context, which centres on the traveler's location and other relevant locations, plays an important role in personalizing PT PIS. Many early studies focus on these aspects, with nearly half of the articles considering real-time data about the **traveler's current location**. This information is essential for delivering services relevant to the traveler's geographical position. GPS signals from users' phones are the most common method for obtaining this data (e.g., [58, 62, 162, 178]). However, determining when precise location data is necessary versus when approximations are sufficient can help reduce complexity, data usage, and battery consumption [30, 152]. Numerous alternatives exist for measuring a person's location [39]. In the sample, these include accelerometers [152], Wi-Fi, or GSM signals [30, 51, 64]. However, relying on signals detected from vehicles can create challenges in distinguishing between passengers inside the vehicle and those nearby.

Research often links traveler location to **other locations** for wayfinding [60] or seat-finding [89], sourcing data from private networks, maps, open datasets, or APIs (see Table 2 in supplementary material). This data typically represents distance to identify nearby stops, stations, or PoIs (e.g., [4, 128, 145]). Vehicle location data also enables real-time updates (e.g., [64, 120]), often via sensors and GPS [75]. However, this faces logistical, connectivity, and cost challenges [75, 116], which researchers may mitigate by balancing real-time data with historical predictions.

4.2.5 User Characteristics. Numerous articles consider attributes related to user characteristics, classifying them into three subcategories: demographics, (dis)abilities and special needs, and knowledge level.

Demographics include inherent traveler attributes like age, gender, and ethnicity (e.g., [4, 34, 44, 46, 124]) and socio-economic factors such as profession, education, and income (e.g.,

[83, 101, 111, 142, 173]). Table 5 shows these attributes are frequently considered in PT PIS requirements and algorithms, such as predicting a traveler's transportation mode or route choice [83, 111]. Demographic data is typically obtained explicitly, during initial system interactions [4, 26, 46, 107] or through questionnaires [83]. Proper handling of this personal data is important; some studies ensure secure storage, normalization, and anonymization to protect privacy [4, 101, 173], though many others fail to discuss this aspect.

In user modeling, simple models based on aggregated demographic attributes such as age groups allow for practical implementations with the available data [142]. While extending these models could enhance recommendations, it requires substantial data resources [142]. Alternatively, demographic attributes can help calculate similarities between users, facilitating recommendations based on shared attributes and interaction histories [44]. However, challenges remain in including implicit information inferred from clustering techniques in the recommendation process, which requires further development and innovation [44]. Despite their use in user models, the contribution of demographic attributes to effective personalization is often unclear. This raises important questions about the necessity of demographic data in personalized PT PIS and whether their potential benefits outweigh the associated privacy risks.

Research approach **(dis)abilities and special needs** attributes broadly, addressing both general (dis)abilities and specific instances, such as the ability to navigate stairs and slopes (e.g., [53, 135]), seating needs (e.g., [1, 145]), and wheelchair accessibility (e.g., [64, 135]). Some studies also account for physical and mental conditions, including visual and hearing impairments (e.g., [34, 124, 132]), cognitive abilities (e.g., [22, 107]), and allergies [4]. Accessibility needs are mostly communicated explicitly by travelers (e.g., [4, 25, 60, 135, 145]). However, some research focuses on pre-determined stereotype user profiles. For example, Pressl et al. [132] developed profiles for various disabilities based on findings from a pre-study, enabling unregistered travelers to select a standard profile. At the same time, registered users can specify individual settings. Similarly, Bravo et al. [19] create disability-specific stereotypes, assuming that individuals with similar disabilities share comparable navigation preferences. These profiles are not static; users can adjust settings to better reflect their needs. Walking speed is another factor, determined via user input [53] or inferred from user behaviour [29, 124]. Personalization incorporating criteria concerning disabilities leads to multi-criteria search problems, which are inherently more complex than single-criterion problems due to ambiguity [132].

The implementation of **knowledge level** as a personalization attribute in PT PIS remains largely unexplored, as evidenced by the TRP. Travelers sometimes provide this data directly, while self-learning digital assistants and external databases can infer it when needed [81]. Familiarity with the PT network is a key component of knowledge level, encompassing experience with specific trips (e.g., [96]), transport modes (e.g., [110, 111]), and interchanges (e.g., [115]). Such familiarity information can be integrated into ML models, including random forest classifiers or multilayer perceptrons, to predict transport mode choices and compute personalized switching gains [83, 111].

A related concept, skill level, reflects a traveler's proficiency with a PT PIS and is estimated through interaction metrics such as the number of prior sessions, dialogue turns, session durations, and the time required to access specific services. Profiles are updated, categorizing travelers into low, medium, high, or expert skill levels. For example, Griol et al. [62] classifies users' abilities in spoken interaction systems, updating profiles at the end of each session. Komatani et al. [92] dynamically uses features such as filled slots, speech recognition scores, and silence durations to classify skill levels. These classifications enable adaptive dialogue strategies that evolve during user interactions.

4.2.6 Ownership. Objects owned by travelers could play an important role in personalization for PT PIS, encompassing three primary subcategories: subscriptions and tickets, private vehicles, and personal items.

Subscriptions and tickets receive limited attention but encompass both booked tickets or reserved seats for a specific journey (e.g., [89, 95]) and subscriptions for set periods, such as monthly or seasonal plans (e.g., [30, 111]). These attributes influence a traveler's mode choice, by determining which modes a pass covers [74]. Subscription plans also reduce the importance of cost in transportation decisions. While subscriptions and tickets can often be automatically linked to user accounts, manual updates remain a common practice [46].

Little research considers the ownership of **private vehicles**. This encompasses ownership of cars and bikes (e.g., [46, 111]) and possession of a driver's license (e.g., [68, 142]). Access to these expands a traveler's options, especially during PT disruptions or for last-mile solutions, such as shared mobility. However, incorporating private vehicles in travel recommendations must balance traveler satisfaction, PT operators' profitability, and environmental impact.

Personal items. These attributes consider personal items unrelated to accessibility needs, such as luggage (e.g., [26, 149]) and strollers (e.g., [19]). Carrying personal items can impact a traveler's flexibility, mobility, and preferences, such as wanting to avoid crowded spaces [177]. A challenge in considering this attribute is its variability, as travelers need to provide this information for each trip where they carry items. Notably, articles in this subcategory were all at lower TRLs (TRP 1 or 2, see Table 5).

4.2.7 User Preferences. User preferences are key to personalization in PT PIS, with studies addressing attributes across seven subcategories: communication, mode, interchanges, PoIs and routes, time and day, walking and cycling distance, and other people. Preferences can be acquired explicitly or implicitly, each with its advantages and challenges. Explicit preferences are provided by travelers through trip-planning options or saved in user profiles. While typically more accurate, this method requires user effort for setup and regular updates. When users neglect updates, systems may rely on default or outdated criteria [157]. Implicit preferences, inferred from user behaviour using ML models, eliminate the need for manual input. This approach reduces setup time and enables personalization even when data is incomplete [44]. However, challenges with implicit methods include cold-start issues, inaccurate assumptions in atypical situations, and increased complexity or reduced interpretability when preferences are modeled across many features [173].

Preferences related to **communication** include language and modality. These can be explicitly defined via app settings [78], device configurations [91], or inferred implicitly through technologies like speech recognition [92]. Updating the user profile after each interaction improves settings for future interactions [62]. While most studies addressed general communication preferences, Zografos et al. [181] also explored preferences for the type of information and device used, noting that needs vary during different journey stages. For instance, written communication is typically more effective before departure, whereas spoken interaction may be more appropriate during travel.

Research often uses journey-related preferences, involving specific attributes or the general preference for **POIs and route**. Users explicitly define favourite places and routes through saved stops or destinations (e.g., [51, 60, 75]), while general route preferences can also be set manually [26]. More recently, research increasingly employed algorithms to learn preferences from historical interactions or route choices. These models incorporated the interaction of multiple attributes related to a preferred route (e.g., [8, 96, 121]) and are often trained using **automatic fare collection** (**AFC**) data. Unfortunately, due to its restricted availability, AFC data typically requires partnerships with transportation operators. To address cold start problems,

systems may leverage endogenous recommendations [150][3] or allow users to manually specify their preferences [19, 115] when there is not enough data on interaction or usage history yet.

Several more specific preferences were identified across personalized PT PIS, often allowing users to manually select them [135] or learn them from historical data [123]. **Mode** preferences often include favoured transportation types, specific vehicles, or operators (e.g., [20, 62, 89, 116]). **Interchanges** preferences spanned aspects like direct routes, maximum allowed transfers, and timing of connections (e.g., [29, 53, 66, 114]). Moreover, some studies specifically focused on preferences concerning mode changes (e.g., [11, 71]). **Walking and cycling** preferences often focused on maximum distance (e.g., [29, 81, 124]) or time (e.g., [8, 66]) for comfort or accessibility.

Some studies explore preferences related to **other people**, particularly travelers' tolerance for crowded environments (e.g., [81, 83]). While crowdedness is often measured objectively, it remains underexplored as a subjective preference. Perceived crowdedness can vary among individuals and contexts [99], influencing tolerance for crowded connections and highlighting the need for further research. A few studies have also examined preferences regarding social interaction with other travelers (e.g., [137, 145]) and assistance from staff (e.g., [1]). Though relatively niche, these preferences are vital for specific demographics like elderly or disabled travelers. However, practical implementation of these preferences remains challenging, as reflected by their classification in lower technology readiness phases (TRP 1). For instance, accurately identifying travelers who are, or plan to be, in the same vehicle is a complex task [137].

Time and day preferences mainly include departure or arrival time, which many PT PIS require travelers to provide explicitly (e.g., [46, 116]). Other preferences include travelling during specific times, like in daylight hours (e.g., [1, 30, 78]) or adhering to time constraints such as appointments (e.g., [88, 135, 158]). While these preferences currently rely on manual input, future systems could integrate with users' calendars to infer time constraints automatically.

Although systems can address preferences individually, their interactions shape the travel experience [166]. Broad factors like *convenience* (time, crowdedness, and seat availability) [11], *comfort* (crowdedness, interchanges, walking distance, unfamiliar surroundings, social status) [81], and *safety* (time of day, light outside, and crowdedness) [1] show this complexity. Personalized PT PIS must balance these elements to provide meaningful, user-centric adaptations (e.g., [1]).

4.2.8 Trip-related Attributes. Trip-related attributes, such as costs, carbon emissions, energy consumption, and health impacts, are fundamental to any PT PIS. However, these attributes were not categorized as either user or context attributes because they are often more overarching, influencing both models rather than fitting exclusively into one. Additionally, they are frequently inherent to the system and not explicitly calculated or distinguished in many studies. Despite this, they remain important. Research commonly uses costs to rank routes (e.g., [8, 87, 101, 112, 123, 145]). Costs are considered an *impact* attribute, defined as a measurable factor quantifying the impact of a route, either on the individual traveler or the broader environment. Impact attributes assess the outcomes and implications of a travel choice. Other impact attributes include carbon emissions (e.g., [44, 140, 144]) and energy consumption [66]. Researchers often estimate these attributes using predictive tools or data provided by transportation operators, although some studies do not specify how they determine these values. Health impacts of a trip (e.g., [1, 110]) are often linked to the transportation mode. For example, walking or cycling distances contribute to the health benefits of a journey, which could be quantified through metrics like calories burned. However, health considerations are rarely primary factors for travelers when selecting routes [46]. Finally, we identify comfort as an impact attribute, often derived from multiple factors [23, 169].

[3]Recommendations designed with the purpose to learn more about unknown criteria and weights.

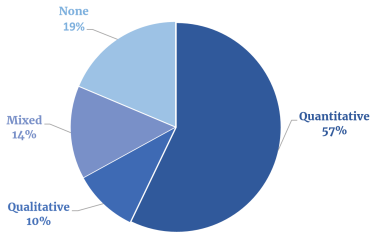

Fig. 3. Percentage of articles with no, quantitative, qualitative, or mixed evaluations.

4.3 Dimension III: Evaluation of Personalized PT PIS

The final dimension of our framework focuses on the evaluation of personalized PT PIS. In our sample, seventeen articles did not include an evaluation (see Table 6), with the vast majority classified in TRP 1 (research). Figure 3 shows the proportion that included no, quantitative, quantitative, or a mix of quantitative and qualitative evaluations.

In our framework, we distinguish three main categories: *evaluation type*, *evaluation metrics*, and *evaluation layers*, further dividing them into subcategories (see Figure 2).

4.3.1 Evaluation Type. In PT PIS evaluation, studies typically fall into five categories: user studies, stakeholder studies, expert analyses, simulations, and data studies. Table 6 shows an overview of each study type count and count per TRP.

User studies gather new data directly from current or potential end-users. Table 7 categorizes the articles by user study type and sample size.[4] The table also shows the number of articles for each user study type per TRP level.

Within user studies, *interviews* are a common method. Researchers often use a semi-structured format, a widely used approach in Human-Computer Interaction research [14]. Here, the main questions are decided upon beforehand, and follow-up questions are asked based on the participants' responses. These interviews typically have small to medium sample sizes (see Table 7), which are generally considered adequate for such qualitative studies. Some studies, such as [110], enhance generalizability by combining interviews with other methods. While most studies conduct interviews in controlled lab settings, some are carried out in the field, such as in PT environments [22] or while driving participants [135]. The data collected from interviews is qualitative, but apart from [110], studies do not report analysis methodologies in a reproducible manner. This highlights the need for greater transparency and consistency in qualitative analysis within PT PIS to improve comparability across studies.

The second type of user study involves *surveys*, where participants answer written questions, sometimes after a demonstration or system interaction. Surveys are primarily used in TRP 1 to understand user needs and preferences [1] and define requirements [132]. However, they are also used to evaluate prototypes or implementations in other TRPs. Most survey results are quantitative, with researchers often using Likert scales to measure usability and usefulness. While valuable, surveys may fail to capture detailed feedback, which can be counteracted using qualitative or mixed-method surveys [1, 132, 144]. Sample sizes vary widely, from small to large (see Table 7). Surveys are mainly conducted online but occasionally in the field, such as at metro stations [83] or airports [181], or in the lab [111].

[4]When an article used independent studies of multiple types, they are reported under each respective type. When an article contains multiple studies of the same type, the sample size of the largest is reported. User tests and field tests often included a short interview or survey afterward; for simplicity, such studies are solely classified as user test or field test.

Thirdly, *observations*, particularly ethnographic methods, are used in a few studies to examine participants' daily lives. For example, Carmien et al. [22] observe high school students with cognitive impairments learning to use buses, while another study explores how bus passengers engage with social media using cultural probes [57]. Although only two studies in our sample used this method, this approach can offer more insights into actual behaviour [14].

A fourth approach, *user tests* involves participants completing well-defined tasks, most commonly in TRP 2 studies. In TRP 1, two studies had participants choose from travel options for given trips, producing data for travel choice prediction [8, 121]. Most studies conducted user tests in labs, sometimes with simulated environments (e.g., a car [143] or tram setup with seating and background noise [158]). In the latter, researchers use a Wizard-of-Oz method, mimicking system behaviour to test a concept before full implementation. Some tests took place in the field, such as testing travel assistance for visually impaired users [60] or pedestrian guidance [135] in a station. Occasionally, tests are conducted online through surveys (e.g., [34, 142, 157]). For instance, one study had users choose weather and climate conditions and then rate recommended routes [34]. Data collected was mostly quantitative, often using surveys for user ratings, though two studies used the observational thinking-aloud method [71, 158].

Lastly, in *field tests*, participants use systems as part of their everyday routines without any assigned tasks. Unsurprisingly, researchers primarily mention field tests in articles classified as TRP 3, although there were exceptions. For instance, a TRP 1 article with a field test for an experimental proof of concept in which participants provided feedback on their trip experience and the recommended route came from Google Maps [11], and a TRP 2 article with a field test in which the ratio of participants' routine versus non-routine routes was measured [114]. This latter field test was clearly in the TRP 1 phase, but the same article also reported a user study with an implemented prototype. Surprisingly, one field test used the Wizard of Oz method to simulate social-media-based travel advice [57].

Overall, while studies employ various user study methods, there is potential to explore additional approaches, such as focus groups or diary studies. Moreover, the incomplete reporting raises concerns about reproducibility and comparisons across studies. Our review also highlights the challenge of creating representative samples in user studies. Quantitative studies require larger samples for statistical validity, whereas qualitative studies depend on generalizability and data saturation. Unfortunately, it is frequently unclear whether studies have reached data saturation. Moreover, ensuring user diversity remains challenging, though our review shows some variety. While many studies focused on generic participants, often passengers in field tests, four articles explicitly address users with special needs. These included visually impaired users [60, 132], wheelchair users [132], high school students with cognitive disabilities such as autism and Down syndrome [22], and older adults [1]. Despite this, age diversity remains limited. Only 12 articles explicitly mentioned participants' age range, and five implicitly indicated it (4x students, 1x older adults). Only five included older adults ($\geq$ 60 years) in the sample [1, 110, 116, 142, 144], a group that could greatly benefit from personalization.

Stakeholder Studies gather insights from people affected by PIS without being end-users. Few studies employ this method (see Table 6), with six classified in TRP 2 or TRP 3. Most aimed at gathering requirements (related to TRP 1), typically after the system had already progressed. Notably, four studies use both stakeholder and user research.

Studies mainly use qualitative methods, including workshops, interviews, observations and participatory design. Workshops identified use cases [43] and validated scenarios where traveler context information could enhance travel support [81]. Interviews allow for a deep understanding of the current state of PIS, including potential risks and benefits of personalized information [80, 81], common navigational problems [22], and requirements [57]. Observations offer insights into the

Table 6. Study Types Performed and Articles without Evaluation, Showing the Total Number of Articles Per Type and the Number Per TRP

Study type	#Articles	TRP1	TRP2	TRP3
User Study	39	[1, 8, 11, 83, 87, 88, 110, 121, 132, 157]	[22, 34, 62, 71, 78, 92, 111, 112, 114, 123, 142, 143, 158]	[17, 46, 57, 58, 60, 64, 75, 101, 104, 115, 116, 133, 135, 144, 178, 181]
Stakeholder	7	[43, 81]	[22, 80]	[57, 133, 181]
Simulation	22	[13, 23, 53, 120]	[5, 19, 25, 44, 62, 65, 67, 111, 114, 128, 150, 169]	[58, 66, 75, 101, 116, 133]
Data Study	21	[18, 35, 38, 83, 95, 96, 108]	[27, 51, 92, 98, 111, 128, 140, 162, 170, 174, 180]	[58, 64, 101]
Expert Analysis	19	[23, 87, 120, 121, 157]	[51, 65, 67, 68, 80, 174]	[4, 20, 101, 116, 135, 152, 178, 181]
No evaluation	17	[30, 33, 89, 91, 103, 107, 124, 137, 145, 149, 173, 177]	[26, 29]	[7, 56, 163]

Table 7. User Studies: Per Study Type Articles That Used it with the Largest Sample Size Used, and the Number of Articles Overall and Per TRP

Type	# Articles: Total and TRP1 TRP2 TRP3	Sample Size: Unknown	Very Small	Small	Medium	Large	Very Large
Interview	5			[22, 64, 116]	[110, 135]		
Survey	11	[1, 133]		[123]	[111, 132]	[87, 88, 181]	[83, 110, 144]
Observation	2		[57]	[22]			
User Test	17	[104]	[22, 62, 114, 121]	[60, 71, 78, 112, 158]	[92, 135, 143, 157]	[142]	[8, 34]
Field Test	13	[58, 75]		[57, 114, 115]	[11, 116]	[17, 46, 144]	[64, 101, 178]

Sample sizes: very small (<10), small (10-19), medium (20-99), large (100-199), and very large (≥200).

working practices of staff managing outwardly facing social media feeds [57], while participatory design helps create appropriate system responses for different scenarios [22]. Some studies also incorporate quantitative approaches, such as surveys to assess cost-effectiveness [181], or mixed methods producing qualitative and quantitative data [133]. Most of these studies conducted their research in the lab,[5] with one survey conducted after system interaction [181]. Though a few interviews and observations took place in the field [22, 57]. Samples show the variety of stakeholders involved in PT PIS, including mobility providers [22, 43], local authorities' staff and PT domain experts [81], caregivers and teachers of special needs students [22], staff managing social media [57], bus drivers [133], and travel information providers [181]. While this diversity of stakeholders offers valuable perspectives, it also presents the challenge of balancing these views and considering their impact on the broader personalized PT PIS ecosystem.

Expert Analyses use experts who are part of the research team to analyze properties of systems, algorithms, or models by (1) using a method that does not require data, such as benchmarking or mathematical proof, (2) researchers taking on the role of study participants themselves, or (3) researchers taking on the role of system engineers.

In *benchmarking studies*, researchers compare the functionality of systems [116, 157] or frameworks [20], whilst *mathematical proofs* demonstrate an algorithm's correctness [23]. These studies are typically conducted in a lab environment and produce quantitative data. Whilst effective for identifying functional differences, benchmarking studies could also benefit from incorporating qualitative insights that explore the "why" behind the numbers, rather than focusing solely on binary feature comparisons.

In *researchers as study participants* studies, researchers typically test system functionality in realistic field conditions. For example, they evaluate systems by driving cars on specific routes [135], walking through cities [4], or taking PT trips [178]. Compared to users, researchers often focus on specific elements, such as the power consumption of location providers [152] or mode

[5]One was conducted online due to COVID restrictions [80].

Table 8. Overview of the Elements Simulated in the Simulation Studies

Main Type	# Articles	Sub Type	# Articles	Refs.
User behaviour	15	Trip requests	8	[23, 44, 53, 66, 67, 75, 101, 116]
		Route selections	4	[65, 120, 128, 150]
		Interactions	3	[62, 114, 150]
		User movements	3	[13, 65, 120]
User preferences	5			[23, 53, 66, 67, 128]
Status	7	Transport and traffic flow	4	[25, 65, 111, 120]
		Specific vehicle movements	3	[58, 133, 169]
Infrastructure	4			[13, 19, 65, 116]

detection [178]. One study tracked disruptions at a station over time to compare alternative route recommendations to other systems' [80], while another assessed the system for different scenarios in the lab, producing qualitative data [68]. In *researchers as system engineers* studies, researchers take on the role of system engineers and modify the system or its underlying models to explore how different components, features, or parameter values influence system performance (e.g., removing one in an ablation study). These studies are conducted in the lab and produce quantitative data.

Data studies use existing data for evaluation. Most studies use user data (15/21 articles) to investigate the accuracy of route recommendations and predictions like activity duration and mode detection. Researchers typically evaluate ML approaches using cross-validation, splitting data into training and test sets. To mitigate bias from data splitting, the wider personalized systems community uses rigorous n-fold cross-validation [125], dividing data into n segments for repeated testing [90], but only four articles adopted this: 10-fold in [92, 95, 96] and 5-fold in [62]. Other studies use status data to assess the accuracy of busyness prediction [58, 64], and real-time schedule data to assess the accuracy of travel duration prediction [18] or best route selection [140].

Data studies are typically summative, reporting quantitative metrics. Hence, they often fail to provide insights into why a system or component performs well or poorly and how to improve it. While comparing alternatives that differ in a single aspect could offer valuable insights, most studies did not test such comparisons. Additionally, diverse datasets from different countries with varying infrastructures, policies, and cultural contexts complicate comparisons. Moreover, the limited reporting on parameter value selection reduces reproducibility and increases the risk of overfitting models.

Simulations generate new data by simulating real-world phenomena, like passenger behaviour or traffic flows, often using existing data. Table 8 summarizes the elements simulated. Most simulations focus on *user behaviour*, particularly trip requests defined by origin, destination, and time of travel. Simulations also modeled route selections and user interactions with interfaces, including clicks and dialogue acts. Moreover, some studies simulate user movements, including traveling to bike-sharing stations and boarding vehicles. However, most simulations did not incorporate real user data, limiting their ecological validity. Some relied on schedules or PoIs to create plausible trip requests.

Simulations of *user preferences* mainly involved generating users with varying preferences regarding journey quality attributes such as journey time, number of transfers, costs, CO2 emissions, and walking distance, alongside simulated trip requests. Another study also simulated users with diverse accessibility preferences and varying risk tolerance levels combined with simulated route selection [128]. These simulations were simple, relying on randomly generated, probability-based, or systematically varied preference vectors. Researchers did not use data for preference simulations, indicating the ecological validity of these studies could be improved by using actual preference distributions from users.

Simulations of user behaviour and preferences are often used over studies with real users because they are faster, less costly, scalable, allow extended testing for adoption, and can easily represent the many different types of users needed to fully measure the impact of adaptation or the accuracy of user modeling [125]. However, as Paramythis et al. [125] note, simulations can be limited by similar assumptions underlying the adaptive system's design, which may be flawed.

In addition to users, *transport and traffic flows* are simulated using schedule, status (vehicle locations, traffic), vehicle attributes, and road segment data, e.g., a metro network with train locations, trains arriving, and platform loads [120].

Specific PT vehicle movements can also be simulated, such as bus movements [133]. Instead of existing datasets, this simulation uses GPS data from a car driving along a route at different times, mimicking bus behaviour [169]. Simulations of vehicle movement, transport and traffic flows enable rapid and safe testing of various scenarios and algorithms.

Lastly, *transport infrastructures* are simulated, ranging from simple to complex models. Simpler simulations often do not rely on datasets, such as creating pedestrian networks with certain sidewalk properties to evaluate generated routes adapted to disabilities [19] or simulating a bike-sharing infrastructure [13]. More complex simulations use a multi-agent approach, simulating stations, road, pedestrian, biking, PT segments, vehicles and travelers [65, 116]. These based their infrastructure simulations partially on schedule and/or road segment data. By simulating transport infrastructures, researchers can rapidly test various contexts under controlled conditions.

Data used in the data studies and simulations. In our sample, researchers use a variety of data in both simulations and data studies (see Table 9). Often, data is publicly available (see Table 2 in supplementary material). Otherwise, data was provided by operators or government agencies (e.g., [18, 56, 116]).

Schedule data. Most studies use the **General Transit Feed Specification (GTFS)**, an open standard for PT data.[6] Generally, studies report data size in terms of the number of PT lines and stops.[7] Most datasets are from large and very large[8] cities (e.g., [18, 44, 65, 67, 116]). Some datasets are smaller: from a large or medium-sized city [23, 51, 116], or a small subset of a city (e.g., one origin/destination pair and related lines [53, 120, 140] or lines from one transport mode [83]). For generalizability, it helps if data studies are repeated for different locations. However, in our sample only three articles use datasets from multiple locations [35, 101, 116]. Only one article used data beyond cities and metropolitan areas [35]. *Real-time* scheduling data was only mentioned in three articles [18, 51, 140].

User data mostly consists of AFC data, originating from smart travel cards such as the London Oyster cards [95, 96]. This data is typically not publicly available, so researchers need connections with transportation agencies to gain access. AFC data provides trip records such as smart card ID, origin/destination pairs, fares, and date and times of checking in/out. However, in some cities (e.g., London and Shanghai), bus users' destinations are unavailable in the AFC data because they pay a flat trip fee and do not check out. Hence, some articles restricted the data used to metro data. As AFC data is very large, articles tended to select a subset, with the size indicated by the time period, the number of users included, and the number of trips. Most evaluations failed to consider how well the system performs in different time periods, whilst travelers' route choices may be impacted by seasons. When restricting the users to include, typically users were included who had travelled much to ensure enough data availability. Hence, performance for less frequent travelers or new travelers (so-called cold-starts) was often ignored.

[6]url: gtfs.org

[7]We use the term "stop" to include bus/tram/metro stops, train stations, and airports.

[8]We classify cities based on population size, using medium for a population 200–500K, large for 500K–1M, and very large for ≥1 million.

Table 9. Overview of the Data Used in Simulations and Data Studies, Including the Number of Articles and Identified Main and Subtypes

Main Type	# Articles	Sub Type	# Articles	Ref.
Schedule	18	Planned	15	[23, 35, 44, 53, 65–67, 75, 83, 101, 111, 114, 116, 120, 128]
		Real-time	3	[18, 51, 140]
User Data	20	AFC	13	[13, 27, 65, 83, 95, 96, 98, 108, 111, 120, 170, 174, 180]
		Interactions	3	[38, 92, 101]
		Survey	3	[83, 111, 150]
		Purchases	1	[95]
		User locomotion	1	[162]
		Unspecified	1	[23]
Status	10	Traffic	4	[25, 58, 111, 116]
		Demand and crowdedness	3	[13, 64, 67]
		Vehicle locations	3	[120, 128, 174]
Other	12	Road segments	9	[18, 35, 44, 53, 101, 111, 116, 162, 174]
		PoIs	4	[66, 101, 162, 174]
		Pavements, crossings, and so on	2	[53, 128]
		Weather records	2	[101, 174]
		Vehicle attributes	1	[65]

In addition to AFC data, other user data provides insights into the trips users were going to make. Compared to AFC data, these datasets are mostly small.[9] Data on user interactions include a corpora of dialogues with PT systems [92], social media posts about PT [38], and travel queries [101]. Survey data came from travel diaries [111, 150] and station-based surveys [83], which gathered 15K trips but are limited to that location and context. Additionally, a small dataset contained users' locomotion data from sensors for only three users [162]. Lastly, a large purchase dataset of London pay-as-you-go and travel cards was used to investigate impact on costs [95].

Status data. This subcategory included data about the status of roads, stations, and vehicles. Real-time traffic information came from traffic performance [25, 111] and traffic control systems [58, 116]. Demand and crowdedness data included bus and metro travel demands [67], crowd density estimation reports (generated from bus WIFI data) [64], and data on the availability of shared bikes and bike slots [13]. **Automated vehicle location** (**AVL**) tracked mass transit vehicles [120, 128], and shared bikes [174].

Other data mostly consists of data on road segments, where the size is in the number of segments. Most road segment datasets are large, from very large cities [18, 101, 111, 116]. Some are smaller: from a large [174] or medium-sized city [116, 162], or a subset from a city [53]. Only one article uses datasets beyond cities and metropolitan areas [35]. PoIs came from map APIs (2 M from Baidu [101]) and GTFS data (only 14 [66]). Lastly, data on the weather (from Baidu), vehicle attributes (capacity, speed), and pavements and pedestrian crossings were used.

4.3.2 Evaluation Metrics. Figure 2 highlights the main categories identified within this dimension of the framework, while Table 12 shows the number of articles per category and sub-category. Additionally, the heat maps in Tables 10 and 11 show per metric what study types were used[10] and per study type what metrics were investigated.[11]

Requirements focus on insights for specifying and designing new systems, including (1) current travel behaviour and related artefacts, (2) opinions and issues with the current situation, and (3) motivations, preferences, and needs for travel and system support. Studies typically gather requirements in early stages, evident from many articles in TRP 1 and higher TRP articles discussing

[9]Exceptions were the travel query and purchase datasets.
[10]Articles may contain multiple study types, so percentages are based on study type instances. Where an article contained multiple studies of the same type, these were counted as one instance.
[11]If an article contained multiple studies of the same type, they were regarded as one.

Table 10. How Metrics Were Investigated: % Metric Type Instances That Were Investigated Using a Particular Study Type

		Study type				
		User Study	Stakeholder Study	Simulation	Data Study	Expert Analysis
Metric Type	Requirements	70%	26%			4%
	Usability	86%	5%	9%		
	Usefulness	59%	2%	25%	7%	7%
	Accuracy	25%		18%	48%	10%
	Technical Performance	24%		35%	6%	35%
	Model/Algorithm Efficacy			29%	24%	47%

Table 11. What Was Investigated Per Study Type: % Articles That Used a Particular Metric for That Study Type

		Metric type					
		Requirements	Usability	Usefulness	Accuracy	Technical performance	Model/Algorithm Efficacy
Study Type	User Study	41%	49%	67%	26%	10%	
	Stakeholder Study	86%	14%	14%			
	Simulation		9%	50%	32%	27%	23%
	Data Study			14%	91%	5%	19%
	Expert Analysis	5%		16%	21%	32%	42%

requirements in their initial phases. Effective system development requires considering all stakeholder perspectives, yet most studies focus solely on travelers via user studies. Fewer consider stakeholders like mobility providers or authorities [43, 80, 81], and only three combine user and stakeholder perspectives [22, 57, 133].

Usability covers a broad spectrum of ease of use: (1) whether the system is easy to use in general, (2) satisfaction (how much users like it), (3) how much users trust it, (4) efficiency (how fast it is to use and how much effort it takes experienced users), (5) learnability (how easily new users learn to use it, including not having to rely on memory), (6) task completion (whether users can complete tasks with few errors), (7) error recovery (how easily they can fix errors), and (8) issues with and improvement suggestions for usability (including system appearance). Evaluating these aspects ensures users can successfully and satisfactorily interact with the system. Studies mainly obtain usability metrics through user studies. These are typically measured through standardized methods such as the **User Experience Questionnaire** (**UEQ**) [71] or the **Computer System Usability Questionnaire** (**CSUQ**) [143]. However, researchers also compose their own measurements (e.g., [92, 181]). In a case where end-users were cognitively disabled, they conducted a stakeholder study [22]. Studies also use simulations to evaluate system performance, such as task completion and efficiency in dialogue systems [62] and route information systems [114], often alongside user studies.

Usefulness reflects how well a system achieves its intended outcomes. This includes (1) the system's impact on (intended) travel behaviour, such as CO2 emissions and fuel consumption, (2) users' usage of and willingness to (re)use the system, (3) user acceptance, (4) trip time reduction, (5) trip quality improvement, (6) cost-effectiveness (in terms of reducing costs for travelers and for other stakeholders), and (7) opinions and issues regarding the system functionality.

The evaluation of usefulness is inherently subjective, as it depends on the stakeholder. For example, users may prioritize trip quality, while operators focus on operational efficiency, and municipalities consider broader societal impacts. Neglecting any perspective can undermine a system's practical feasibility and acceptance. Therefore, usefulness is ideally evaluated from multiple perspectives. A notable example is provided by Hao et al. [65], who consider the perspectives of travelers, transport operators, and the government. However, most studies only consider the traveler

perspective, often relying on user studies. One article uses a stakeholder study [181] which considers travel information providers' evaluation of a system's socioeconomic impact, considering costs for design, training, and operation [181].

User- or stakeholder studies are suitable for evaluating subjective usefulness. Usefulness metrics such as trip time, trip quality, traffic flow, congestion and fuel consumption can alternatively be evaluated through simulations (e.g., [23, 58, 65, 75, 133]). Data studies assess the system's performance regarding its ability to deliver fast and reliable routes [170], to support cost savings [95], and to facilitate finding parking spots [174]. Lastly, expert analyses compare a new system or framework's functionality [20, 116] and recommended trip feasibility [80] to that of existing ones.

Accuracy focusses on the correctness of (1) detection (activity/mode, location), (2) models (user, traffic infrastructure), (3) predictions of choices (transport mode, destination), times (travel/activity duration, arrival), busyness (demand for PT, amount of traffic, crowdedness of vehicles and stations, availability of parking spaces), and locations, and (4) route recommendation and information provision. Most studies obtain accuracy metrics from data and user studies. The 19 articles that use data studies overall investigated each sub-metric except the accuracy of information provision and the transport infrastructure model. In contrast, the 10 articles that use user studies only focus on the accuracy of transport mode choice prediction [8, 46, 112, 121], route recommendation [34, 71, 112], duration prediction [116], and information provision [57, 178, 181]. Seven articles use simulations, primarily for the accuracy of route recommendations: whether they matched user profiles [44, 128], eco-friendly recommendations had indeed less CO2 emissions [116], and how less-walking/fast-trip recommendations compared to those provided by other systems [53]. Others include the accuracy of: demand and crowding predictions (e.g., via simulating passenger response to crowding information [120]), estimated arrival time based on simulating dynamic bus movement [169], the traffic infrastructure model (e.g., agents matching road segments [116]), and transport mode choice prediction [150]. Lastly, four articles use expert analyses for the accuracy of different location providers [51], route recommendations [23, 68], and activity and mode detection.

While accuracy is clearly valuable for evaluating objective factors such as eco-friendly routes, shortest paths, or congestion levels, its relevance is less straightforward for subjective experiences or preferences. For instance, a crowdedness prediction might be technically accurate but fail to reflect how a traveler perceives the situation [99]. Likewise, small improvements in route choice prediction accuracy might have a limited impact on user satisfaction. Therefore, accuracy metrics should be paired with investigations into the subjective aspects of traveler experiences.

Technical performance centres on response time, reliability, and resource usage, critical for deployment. For example, a system that is too slow or consumes excessive resources (power, CPU, and internet data) may struggle to meet real-world demands, resulting in poor user experience or operational inefficiency. Studies mainly obtain these metrics from expert analyses, simulations, and user studies. The six articles that used an expert analysis together investigated each sub-metric of technical performance. In contrast, user studies only focused on reliability [17, 135, 143] and response time [17, 34]. Simulations [5, 19, 25, 65, 67, 101] and data studies [35] focussed only on response time. Thus, while the expert evaluations allowed for a more comprehensive evaluation of the technical performance, user studies and simulations focused on one specific technical performance metric. This suggests that, although current research provides a good starting point, it may lack a comprehensive view of how systems perform under varied real-world conditions.

Model/Algorithm efficacy centres on (1) the impact of system components (e.g., via ablation studies [67]), features, and parameter values (cf. parameter sensitivity), (2) training/initialization time and convergence (number of parameter upgrades for good results), (3) scalability (performance with growing data/features), and (4) model properties (feature independence, robustness, and consistency). Most studies obtain metrics from expert analyses, but also from simulations and

Table 12. Metrics Used in the Sample of Articles, Per Metric (Sub) Type

Metric type	# Articles	Metric subtype	# Articles	References
Requirements	19	Motivation, wishes, preferences, needs	14	[1, 43, 46, 57, 80, 81, 87, 110, 124, 132, 144, 157, 158, 181]
		Behaviour, artefacts used	8	[22, 57, 83, 110, 114, 115, 133, 144]
		Opinions, issues existing situation	3	[22, 133, 144]
Usability	21	Ease of use in general	10	[58, 62, 64, 71, 78, 92, 104, 116, 143, 181]
		Issues and improvements	8	[17, 22, 34, 58, 60, 64, 104, 135]
		Understandability, transparency	7	[62, 71, 88, 92, 116, 143, 158, 181]
		Satisfaction	7	[60, 71, 78, 101, 123, 142, 143]
		Efficiency	7	[60, 62, 71, 78, 92, 114, 143]
		Task completion	4	[22, 60, 62, 143]
		Learnability	4	[60, 78, 143, 181]
		Trust	3	[62, 71, 123]
		Error recovery	2	[92, 143]
Usefulness	39	Functionality and issues	11	[17, 20, 57, 58, 60, 88, 104, 116, 135, 143, 178]
		Impact on (intended) travel behaviour, CO2, fuel consumption, congestion	10	[25, 57, 58, 64, 65, 71, 111, 116, 144, 178]
		Usage and willingness to (re)use	9	[57, 62, 64, 71, 92, 101, 115, 116, 181]
		User acceptance	9	[25, 34, 46, 71, 111, 116, 123, 142, 158]
		Trip quality	9	[11, 13, 23, 65, 66, 80, 120, 128, 174]
		Trip time	8	[13, 23, 25, 60, 65, 75, 111, 133]
		Costs	3	[25, 65, 181]
Accuracy	37	Route recommendations	14	[23, 34, 38, 44, 53, 68, 71, 98, 112, 116, 128, 140, 162, 170]
		Prediction: Transport mode choice	8	[8, 46, 83, 101, 111, 112, 121, 150]
		Prediction: Travel/activity duration	5	[18, 96, 108, 116, 170]
		Prediction: Busyness	4	[58, 64, 120, 174]
		Information provision	3	[57, 178, 181]
		Prediction: Destination	2	[108, 180]
		Prediction: Time of arrival	2	[51, 169]
		Prediction: Location	3	[51, 64, 162]
		Model: User	2	[92, 95]
		Detection: Activity/mode	2	[27, 178]
		Detection: Location	1	[51]
		Model: Trafic infrastructure	1	[116]
Technical performance	16	Response time	13	[5, 17, 19, 23, 25, 34, 35, 51, 65, 67, 101, 135, 181]
		Reliability	5	[4, 17, 135, 143, 181]
		Usage: Power	2	[51, 152]
		Usage: CPU	1	[152]
		Usage: Internet data	1	[152]
Model/ algorithm efficacy	14	Feature/alg. component importance	6	[65, 67, 101, 120, 128, 174]
		Scalability	5	[5, 23, 101, 116, 121]
		Training/initiation time	4	[35, 65, 67, 116]
		Parameter sensitivity	4	[67, 87, 101, 174]
		Convergence and diversity	3	[65, 66, 121]
		Robustness and consistency	2	[65, 101]
		Feature independence	1	[83]

data studies. Expert analyses [23, 65, 67, 87, 101, 120, 121, 174] together investigated each sub-metric except feature independence. Data studies examine training/initiation time [35], feature independence [83], feature importance [128], and robustness across subgroups (e.g., gender) [101]. Lastly, simulations focus on training/initiation time [67, 116], scalability [5, 101, 116], and convergence and diversity in evolutionary algorithms [66].

In the metrics used, there is no explicit attention to privacy. Though, perceived privacy as part of trust [71], users' willingness to disclose information [57, 158] and pseudonymization [158] were studied. Stakeholder studies addressed the perceived risks of personalized context-based information which centred around privacy [81]. Other articles noted privacy as important (and often also safety and security) but did not evaluate it, citing it as a system requirement [11, 22, 58], a system rationale [133], a design rationale [4, 64, 95, 115, 180], an issue [142], or a future research need [5, 68]. Given personalization's privacy implications, this aspect warrants greater evaluation.

4.3.3 Layers Evaluated. Evaluating personalized systems is widely recognized as more complex and significantly different from evaluating non-personalized systems (see e.g., [161]). Paramythis et al. [125] proposed a frequently used layered evaluation framework for personalized systems,

decomposing adaptation into layers, each assessed individually along with the overall system. They identify five layers, each with related evaluation goals.

The **collect input data** layer focuses on data acquisition. Key questions are whether the acquisition works well and whether data is accurately captured. Studies examined, e.g., the accuracy of location detection [51] and technical aspects like power, CPU, and internet usage [152]. Ideally, all data sources and technical performance are evaluated.

The **interpret data** layer focuses on the interpretation of input data. Here, the question is whether the system's interpretations align with what is actually happening. In the sample, studies often evaluated the accuracy of activity and mode detection, and of duration, destination, location, and busyness predictions (see Table 12).

The **model the state of the world** layer focuses on whether the system's model reflects the real world. This includes evaluating whether the model is accurate and suitably constructed for the intended application. Studies assessed accuracy in user models, traffic infrastructure models, transport mode predictions, and model efficacy (see Table 12).

The **decide upon adaptation** layer focuses on the adaptation used by the system. At this stage, evaluators ask whether the adaptations are good and appropriate for the user and their context. Evaluations included the accuracy of route recommendations and information provision, and the efficacy of adaptation algorithms (see Table 12).

Lastly, the **apply adaptation** layer focuses on how adaptations are presented (e.g., route suggestions as lists or ratings). Questions are whether users understand and like them. This layer received no attention; while explanations [88] and pseudonymization symbols [158] were studied, neither assessed the implementation of adaptation.

Many studies focus on evaluating a single layer—mainly for accuracy, performance, or efficacy—while comprehensive multi-layer evaluations are often missing. Whole-system evaluations typically overlook individual layers. For instance, studies on requirements and usefulness focus on overall functionality, while usability studies (e.g., [17, 116, 143]) neglect specific layers like the apply adaptation layer. This oversight can miss issues in one layer that impact others. Evaluating each layer separately is resource-intensive and impractical for complex systems. Here, combining layers for evaluation is an alternative [125].

5 How to Use the Framework?

In the preceding sections, we introduced the personalized PT PIS framework, providing researchers and practitioners with a standardized approach for designing, developing, and evaluating personalized PT PIS. Its primary objective is to enhance comprehensibility, reproducibility, and transparency in the field. Each dimension must be addressed thoroughly, leveraging insights from our review to inform and guide decision-making at every stage.

To illustrate the framework's application, we present a fictional case study following Zangerle and Bauer [175]. The context is a PT operator in a large metropolitan area collaborating with researchers to develop a personalized PT PIS tailored to individual traveler needs, improving the experience during service disruptions.

5.1 Defining the Personalization Object

A key decision in creating a personalized PT PIS is determining what to personalize, as this directly impacts data needs, system type, and evaluation methods. The choice should align with the PIS's objectives and travelers' needs. When providing recommendations, consider both what will be recommended and the rationale. For example, recommendations can reduce effort and enhance efficiency by offering optimized routes (e.g., [8, 68]) and improve the travel experience (e.g., [44, 67]), encouraging PT use. Personalized information can enhance PT's accessibility and usability,

supporting on-route assistance for a broader group of users (e.g., [145, 152]), and allow users to make better-informed decisions [158]. Personalized interactions or vehicle settings, though explored less, can enhance the user experience by improving trust in the system and providing quick access to relevant communication [89]. Future research exploring personalizing other aspects requires a clear rationale for the aspects and their alignment with system goals and traveler needs.

Example application. Insights from a study with travelers revealed that while disruption information is clear, many struggle to determine the next best action. Based on these findings, the team focuses on recommending route alternatives as the personalization object. During the planning phase, the system can advise travelers on whether to take an alternative route, wait, find other transport options, or not travel at all. For those already mid-journey, it prioritizes recommending the most optimal alternative to reach the destination.

5.2 Defining the Personalization Attributes

Once the personalization object has been defined, the next step is to select the attributes on which the personalization will be based. The available data needs to be evaluated carefully, considering its added value, reliability, and feasibility for acquisition. Both user and contextual attributes can be used for personalization, with context offering the advantage of better adaptation to travelers' situations (e.g., [91]). On the other hand, user factors can improve user satisfaction and drive continued use of the system (e.g., [116]). This approach allows the service to better align with personal needs and preferences while capturing the diversity within the target users (e.g., [1]). Both attribute types require careful consideration of how data is acquired, whether explicitly through user input or implicitly via, e.g., sensors. It is also crucial to assess if data collection is necessary for the system's goals, especially regarding privacy and security concerns (e.g., [66, 158]). Furthermore, ontologies can help structure and standardize context data [89, 107]. A multi-criteria approach that combines static and dynamic factors is also advisable to achieve a robust personalization strategy [7, 68].

Example application. For the case at hand, the team uses both contextual and user attributes for personalization. The traveler's location is important to determine applicable alternatives. In addition, crowdedness is considered to improve comfort and manage passenger flow distribution across the network. For location, the team abstracts the raw data into high-level categories, such as "home", "at station X", or "onboard a vehicle", which are derived from network signals (e.g., GPS or proximity to stations). However, due to technical limitations and privacy concerns, real-time crowdedness metrics based on sensors or cameras are not feasible. Instead, the team decides to use historical data and user feedback to develop predictive models for crowdedness, ensuring that their recommendations are based on realistic scenarios without violating privacy. When it comes to user preferences, the team acknowledges that travelers' preferences may vary during disruptions. To accommodate this, they choose to explicitly ask users for input during disruptions, ensuring that the system accounts for the traveler's current needs and provides alternatives accordingly.

5.3 Evaluate the Personalization

The framework offers a more standardized approach to evaluating personalized PT PIS, guiding practitioners through the process. The first step is defining the evaluation goal, identifying the evaluation metrics and selecting the most suitable study design. In this process, it is crucial to assess the system's adaptation layers for effective evaluation [125]. For example, if the evaluation focuses on assessing the quality of the raw input data or whether the input data is correctly interpreted, data studies can evaluate its accuracy and efficacy (e.g., [35, 64, 111]). If the evaluation aims at evaluating the implementation of the adaptation in terms of its usability and usefulness, user studies (e.g., [11, 60, 104]) are more appropriate. In all cases, it is important to carefully consider the evaluation method, sample size, location, and analysis approach. Each evaluation method has

its advantages and limitations, and the choice should align with the evaluation goals. The generalizability of results, confounding variables, and ethical concerns (e.g., privacy or data security) also need to be considered. Data triangulation helps to ensure the validity and reliability of the findings. Ultimately, the data gathered from the different evaluation methods contributes valuable insights for system improvement.

Example application. In this case, the team adopts a layered evaluation approach. After evaluating the quality of the gathered data, they develop a choice utility model and test its accuracy using the city's AFC data. They also simulate traveler scenarios to assess the model's performance over time. Next, the team integrates the model into a beta version of the PIS. To assess the impact of personalization on the traveler experience, they conduct a field study involving 1,000 participants over the course of one month. During disruptions, participants are asked to rate their satisfaction with the personalized recommendations through an in-app survey. At the end, a more comprehensive survey is sent to all participants and some are invited for an interview to gather in-depth feedback. This evaluation allows the team to quantify the added value of personalization and identify opportunities for further improvements.

This example illustrates how our framework supports decision-making in developing personalized PT PIS. While the case presented here is simplified for clarity, the detailed review above provides further insights into the process. Furthermore, although the steps are presented in a linear manner for conciseness, it is important to recognize that the actual development process is iterative. Continuous refinement and alignment between the personalization objects, attributes, and evaluation are needed to ensure the system's effectiveness and adaptability over time.

6 Discussion, Conclusion, and Future Directions

This review systematically analysed and organised existing literature on personalization in PT PIS, providing a comprehensive, high-level overview. Moreover, we provided a framework to guide researchers and practitioners by highlighting key aspects they should consider when designing such systems. In this section, we reflect on the results, discuss potential future research directions, and describe the limitations.

Personalization in PT PIS is growing, but research remains in the early stages. Our review reveals a growing interest in personalized PIS over the past decade, particularly in countries with robust PT infrastructures, such as Germany and China. This trend shows the increasing importance of advanced information systems that support travelers throughout their journeys. Researchers emphasise the need to integrate multiple transportation modes to help users navigate multimodal journeys and improve the flexibility and convenience of PT. While most studies focus on buses, metros, trains, and trams, there has been a noticeable rise in research on shared mobility. Half of the articles on this were published between 2020 and 2024, reflecting its growing significance. Reflecting on the TRP, a quarter of the articles are in the deployment phase, while most are still in the research phase. Recent publications predominantly fall into the research or development stages, indicating that personalized PT PIS remain an area of early-stage research. However, it is important to note that scientific publications may not fully represent industry deployment.

Adapting beyond basic journey attributes and offering insight into the "why?" needs further investigation. Our review found that most studies focus on personalizing route recommendations, with less attention to recommending other aspects relevant to travelers. Similarly, research on personalized information primarily addresses basic journey qualities. While this is a good starting point to reduce dissatisfaction during PT journeys, there is potential to offer a broader range of recommendations and information to enhance the overall experience [165]. Moreover, most articles focussing on personalized recommendations, do not offer explanations to travelers. Given the importance of transparency in fostering trust, satisfaction, and convenience [155],

future research should aim at clarifying why routes are recommended [71]. Regarding personalized interaction, our review found that studies on content representation were mostly published before 2018, covering aspects such as dialogue style, alert intensity, and persuasive strategies. While features like tone of voice, detail level, colour schemes, and visualizations offer opportunities to enhance user experience, only one study addressed adapting presentation. Moreover, most articles in our sample focus on the personalization technology, with little attention to the interface design implications from a user perspective. Research on how to present (recommended) content or explanations to users (e.g., [93, 105, 146]), shows potential for further exploration in personalized PT PIS.

Further exploration of personalization attributes and promoting open science. Our review shows that most attributes used for personalized PT PIS relate to easily accessible data like current time, journey timing, location, and routes. These attributes provide the first step in personalization by offering real-time information adapted to a user's current location. In contrast, less commonly used attributes include day of the week, events, communication preferences, interchange preferences, travel companions, knowledge level, and ownership. The under-use of these attributes likely stems from data acquisition and processing challenges. For example, user preferences often require explicit input or inference through advanced algorithms requiring comprehensive data [114]. Additionally, ensuring adequate infrastructure, storage capacity, integration, privacy, and security remains a challenge for data acquisition and processing [59, 158]. These less commonly used attributes are primarily explored in TRP 1, suggesting they remain in the research phase and are not yet implemented in practice. Advancing personalized PT PIS requires investigating these attributes' relevance and potential through a human-centred approach, offering insights into how they influence travel decisions and experiences, and ensuring efforts are spent on implementing the most valuable ones [166].

Moreover, relevant data like travel history, usage patterns, or preferences, is often accessible only to researchers working with PT operators. This restricts replication, validation, and further advancements. Enhancing data accessibility and promoting open science could drive innovation in personalized PT PIS.

Greater focus on comprehensive evaluation of personalized PT PIS. Section 4.3 showed a wide variety of methods and metrics for assessing personalized PT PIS, although some studies had limited evaluations. This overview aims at inspiring more thorough evaluations, ideally with a layered approach. Despite the diverse metrics used, some key metrics for evaluating personalized systems were lacking, including privacy and security, scrutability, user control, explainability, and fairness of user models and adaptation processes [70]. Additionally, there are other evaluation methods which were not used. User studies for personalized systems are challenging due to the time needed for adaptation [125]. Indirect methods, such as the User-as-Wizard approach [106], can address this. Moreover, stakeholder input, present in only seven studies, is crucial for successful personalized PT PIS, highlighting the need for further stakeholder involvement.

Promote attention to accessibility and inclusion. It is encouraging that quite some articles in our sample focus on accessibility, providing personalized recommendations or support for those with special needs. Moreover, several papers demonstrate how thorough evaluations can be conducted with special user groups. This aligns with efforts to bridge the digital divide—the gap between those who have access to technology and those without—by making PT more accessible and inclusive [141, 148]. With about 16% of the global population facing some form of disability [172], and personal mobility being a fundamental human right [50], ensuring accessible and inclusive access to PT is crucial. Future research should continue promoting inclusion and independence in PT, enabling access to essential services such as education, healthcare, employment, and social activities.

More standardized reporting practices. Good reporting practices are crucial for research reproducibility, comprehensibility and transparency. However, interdisciplinary research can lead to inconsistent reporting [164], complicating comparisons. Many articles in our review provided only partial information on personalization goals, attributes used for personalization, and how these were collected. Evaluation methods often lacked clarity on participant type, sample size, metrics, procedures, and analysis. Future research should establish a shared understanding of reporting practices in personalization for PT PIS.

We adopted a systematic approach to search and select literature, though limitations remain. To minimize omissions, we iteratively refined a comprehensive search string with specific keywords and synonyms, and searched multiple databases, including domain-specific ones, for thorough coverage. Data extraction posed challenges as relevant information was often implicit. We addressed this by inferring from text, tables, figures, datasets, and PT provider information. As highlighted in our discussion, improved, standardized reporting practices are needed to address this challenge. Increased transparency will foster better analysis of findings in future publications. All authors participated in data extraction to ensure consensus.

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